Quantum Cognition Machine Learning for Financial Regime Detection:
A Geometric Observable Approach

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Abstract

Quantum Cognition Machine Learning (QCML) encodes data as quantum states in a Hilbert space and represents features as Hermitian operators, inducing a rich geometric and topological structure over the data manifold. While QCML has been applied to intrinsic dimension estimation, financial forecasting, and supervised similarity learning, extracting geometric observables for time-series regime detection remains unexplored. We construct an error Hamiltonian from financial features, compute the quasi-coherent ground states, and extract three scalar observables from the induced quantum geometry: the Berry phase rate, the quantum Fisher information (QFI) determinant, and a multi-lag fidelity measure. In a comparison of 17 methods across 12 historical crises, these QCML observables—fully unsupervised—achieve effect sizes competitive with a supervised Random Forest baseline ($d \approx 0.93$ vs. $d = 1.13$) and substantially outperform standard unsupervised methods (BOCPD $d = 0.59$, Isolation Forest $d = 0.46$). Our central finding is *complementarity*: QCML observables excel on structurally novel crises where supervised methods lack training precedent, and a hybrid ensemble nearly doubles out-of-sample performance ($d = 0.60$ vs. $d = 0.31$). Across five equity ETFs, Multi-Lag Fidelity achieves $d = 1.44$ vs. RF $d = 0.88$ (Friedman $p < 0.0001$), demonstrating cross-asset generalization.

Keywords: QCML, quantum cognition, regime detection, Berry curvature, quantum metric, Fisher information, financial crises, geometric observables.

1 Introduction

1.1 Background and Motivation

Quantum Cognition Machine Learning (QCML) is a framework in which data points are represented as quantum states—unit vectors in a finite-dimensional Hilbert space—and features or variables are represented by Hermitian operators acting on those states [1, 3]. The key innovation is that observables are operators rather than numbers: the expected value of a Hermitian operator A on a state $|\psi\rangle$ is $\langle\psi|A|\psi\rangle$, a number with uncertainty around it, and the non-commutativity of operators encodes context-dependent relationships among features [3]. This "quantum encoding" enables QCML to learn broad abstractions from small amounts of data, achieving a logarithmic economy of representation that overcomes the curse of dimensionality inherent in classical statistical approaches [1, 3].

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises, and the ability to identify transitions in real time has implications for risk management, portfolio construction, and hedging. Classical approaches include Hidden Markov Models (HMMs) [9], Bayesian Online Changepoint Detection (BOCPD) [10], and supervised classifiers such as Random Forests. While effective in specific settings, these methods operate in flat Euclidean feature spaces and struggle with structurally novel crises that differ from training data.

Our central idea is that QCML induces quantum geometry over the data manifold, and that this geometry changes when market regimes shift. Specifically, the error Hamiltonian of QCML [1] defines a smooth map from feature space to projective Hilbert space $\mathbb{CP}^{n-1}$, endowing the data with a Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. These geometric objects provide a family of scalar observables that respond to structural changes in the market state space without requiring labeled crisis data.

1.2 Prior Work

QCML was introduced by Musaelian et al. [3] as a practical machine learning framework inspired by quantum cognition, the hypothesis that human decision-making follows quantum probability rather than classical probability [11]. Candelori et al. [1] demonstrated that QCML encodes data as quantum geometry, showing that the quantum metric tensor captures the intrinsic dimension of datasets robustly even under noise. Abanov et al. [2] developed the mathematical theory of this quantum geometry, including Berry curvature, Chern numbers, and the matrix Laplacian, providing the formal tools we employ.

In the financial domain, Samson et al. [4] applied QCML to financial forecasting, demonstrating its ability to accommodate large numbers of features with logarithmic scaling. Rosaler et al. [5] applied QCML to supervised similarity learning for corporate bonds, showing advantages over tree-based methods in high-yield markets. Samson et al. [6] extended QCML similarity to firm linkages, constructing profitable momentum spillover strategies. Di Caro et al. [7] applied QCML to biomedical prediction, demonstrating superior performance in high-dimensional, low-sample-size settings.

Classical regime detection originates with Hamilton's Markov-switching autoregression [9], and Adams and MacKay [10] introduced BOCPD for online changepoint detection. Supervised classifiers achieve high in-sample accuracy but require labeled data and are prone to overfitting on rare events.

The gap we address is that no prior work has extracted geometric observables from QCML-induced quantum geometry for time-series regime detection.

1.3 Problem Statement and Goals

Research question. Can geometric observables derived from QCML-induced quantum geometry detect financial regime transitions without labeled data?

Objectives.

- **O1.** Construct a Riemannian geometry over financial time series via the QCML error Hamiltonian framework.
- **O2.** Derive scalar observables (Berry curvature rate, QFI determinant, multi-lag fidelity) from this geometry.
- **O3.** Validate against classical and supervised baselines on 12 historical crises across multiple assets.

1.4 Contributions

1. **First application of QCML geometric observables to financial regime detection.** We extract Berry curvature, quantum metric, and spectral gap from QCML-induced geometry and show they carry rich information about market regime structure.

2. **Three novel observables.** Berry Phase Rate, QFI Determinant, and Multi-Lag Fidelity probe distinct geometric aspects of regime transitions (curvature change, manifold volume, state stability).

3. **Empirical evidence of complementarity.** QCML observables and supervised methods detect structurally different crisis types; a hybrid ensemble nearly doubles out-of-sample performance.

4. **Cross-asset generalization.** QCML observables generalize across five equity ETFs, with Multi-Lag Fidelity significantly outperforming Random Forest ($d = 1.44$ vs. $d = 0.88$, Friedman $p < 0.0001$).

2 Theoretical Framework

2.1 Quantum Foundations in QCML

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$ [1, 2]. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$ with $\langle\psi|\psi\rangle = 1$, and an *observable* is a Hermitian operator $A = A^\dagger$ acting on $\mathcal{H}$. The expectation value $\langle A\rangle = \langle\psi|A|\psi\rangle$ is a real number, but unlike classical statistics, the outcome carries intrinsic uncertainty governed by the operator's spectrum.

Given a data point $x = (x_1, \ldots, x_d) \in \mathbb{R}^d$ and a set of learned Hermitian operators $\{A_k\}_{k=1}^d$, the QCML *error Hamiltonian* [1, 3] is:

$$H(x) = \frac{1}{2}\sum_{k=1}^{d}(A_k - x_k\, I)^2\,, \tag{1}$$

where I is the identity. The Hamiltonian $H(x)$ is Hermitian and positive semi-definite, measuring the collective "error" between the operators $\{A_k\}$ and the target values $\{x_k\}$.

Definition 1 (Quasi-coherent state [1]). *The* quasi-coherent state *at x is the ground state of $H(x)$:*

$$|\psi(x)\rangle = \underset{|\phi\rangle,\, \|\phi\|=1}{\arg\min}\, \langle\phi|H(x)|\phi\rangle\,. \tag{2}$$

The map $x \mapsto |\psi(x)\rangle$ is a smooth embedding $\mathbb{R}^d \hookrightarrow \mathbb{CP}^{n-1}$ wherever the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ [1, 2]. This QCML embedding lifts each data point into projective Hilbert space, endowing the data manifold with quantum geometry.

The pullback of the Fubini–Study metric [14] to $\mathbb{R}^d$ defines the *quantum metric tensor* (the quantum analogue of the classical Fisher information metric [15]):

$$g_{ab}(x) = \mathrm{Re}\,\langle\partial_a\psi|\partial_b\psi\rangle - \langle\partial_a\psi|\psi\rangle\langle\psi|\partial_b\psi\rangle\,, \tag{3}$$

and the *Berry curvature* [2, 8]:

$$F_{ab}(x) = -2\,\mathrm{Im}\,\langle\partial_a\psi|\partial_b\psi\rangle\,. \tag{4}$$

Together, g_{ab} and F_{ab} form the real and imaginary parts of the quantum geometric tensor, the central mathematical object of QCML geometry [2]. Candelori et al. [1] showed that the rank of g_{ab} at a point x equals the local intrinsic dimension of the data manifold, and that the spectral gap of g_{ab} robustly estimates global intrinsic dimension even under noise.

2.2 Quantum Cognition and QCML

The intellectual foundation of QCML lies in quantum cognition: the hypothesis that states of knowledge are naturally represented as vectors in a Hilbert space, with questions and observables as operators [3, 11]. Unlike classical probability, where concepts are represented by sets and the complexity of their description grows exponentially with the number of features, quantum representations achieve a logarithmic economy—the number of parameters in a QCML model scales quadratically with the number of features, not exponentially [3].

This has practical consequences. Classical machine learning requires exponentially more data as features increase (the curse of dimensionality). QCML, by encoding features as non-commuting operators rather than as coordinates in a probability simplex, captures interactions between features in the off-diagonal structure of the Hermitian operators without explicitly enumerating all combinations [1, 3]. The "uncertainty" inherent in quantum representations is not a bug but a feature: it enables abstraction, generalization, and the formation of concepts from limited data [3].

Training a QCML model involves learning the operators $\{A_k\}$ by iteratively computing the ground state $|\psi(x)\rangle$ of the error Hamiltonian and updating A_k via gradient descent to reduce the ground state energy [3, 4].

2.3 QCML Perspective on Regime Detection

In our application, financial features are embedded as Hermitian operators in a Hilbert space via the error Hamiltonian (1). The resulting quasi-coherent states $|\psi(x_t)\rangle$ trace a path through $\mathbb{CP}^{n-1}$ as market conditions evolve. We interpret regime detection through the lens of QCML geometry:

- A **regime** is a region of feature space where the quantum geometric properties (metric, curvature, spectral gap) are approximately stationary.
- A **regime transition** is a geometric phase change: Berry curvature spikes, the spectral gap collapses, or state fidelity drops rapidly—analogous to quantum phase transitions in condensed matter [13].

The following formal results establish the mathematical foundations.

Theorem 2 (Smoothness of the QCML embedding)**.** *Let $\{A_k\}_{k=1}^d$ be Hermitian operators and let $H(x)$ be the error Hamiltonian* (1)*. If the spectral gap $\Delta(x) > 0$ for all x in an open set $U \subset \mathbb{R}^d$, then:*

1. *The ground state map $x \mapsto |\psi(x)\rangle \in \mathbb{CP}^{n-1}$ is C^∞ on U.*
2. *The quantum metric tensor $g_{ab}(x)$ and Berry curvature $F_{ab}(x)$ are C^∞ on U.*
3. *All geometric observables derived from g_{ab} and F_{ab} are C^∞ on U.*

Proof. By the implicit function theorem applied to the eigenvalue equation; see Appendix A.

Theorem 3 (Chern number quantization [2])**.** *Let $S \subset \mathbb{R}^d$ be a closed, oriented 2-surface over which $\Delta > 0$. Then the first Chern number*

$$C_1 = \frac{1}{2\pi} \oint_S F_{ab}\, dx^a \wedge dx^b \;\in\; \mathbb{Z} \tag{5}$$

is an integer, invariant under smooth deformations preserving $\Delta > 0$.

Proof. Chern–Weil theorem applied to the $U(1)$ line bundle; see Appendix A.

Important caveat. In our empirical application, we compute curvature integrals over *rolling windows*—open subsets of parameter space—yielding non-integer values. The topological protection argument (noise cannot change an integer) does not strictly apply. Nevertheless, the magnitude of these curvature integrals is empirically informative, and sharp changes between windows are associated with regime boundaries.

Theorem 4 (Curvature–gap bound)**.** *For any pair of indices* (a, b) *with* $a \neq b$*:*

$$|F_{ab}(x)| \;\leq\; \frac{K}{\Delta(x)^2}\,, \tag{6}$$

where K *depends on the Hamiltonian derivatives but not on the gap.*

This bound formalizes the detection mechanism: as the spectral gap $\Delta \to 0$ at a regime boundary, Berry curvature can diverge as Δ^{-2}, producing the spikes our detectors identify. See Appendix A for the proof.

Proposition 5 (QFI volume interpretation)**.** *The pseudo-determinant* $\det^+(g) = \prod_{\lambda_i > \epsilon} \lambda_i$ *equals the squared Riemannian volume element of the non-degenerate subspace of the quantum metric. Thus:*

1. $\det^+(g) \to 0$ *indicates dimensional collapse (approaching a phase boundary).*
2. *Large* $\det^+(g)$ *indicates the state is exploring a high-dimensional region of state space (stable regime).*

Proof. Direct from the relationship between the metric determinant and Riemannian volume; see Appendix A.

3 Model and Methods

3.1 Model Description

Inputs. Daily OHLCV data for equity ETFs are transformed into five normalized features (log returns, 20-day rolling volatility, z-scored volume, relative strength, momentum), reduced to $p = 15$ principal components, and L^2-normalized.

QCML embedding. The p-dimensional feature vector x_t at each time step is fed to the error Hamiltonian (1) with $n \times n$ Hermitian operators $\{A_k\}_{k=1}^{p}$ ($n = 8$ by default). The ground state $|\psi(x_t)\rangle$ is computed via eigendecomposition of $H(x_t)$.

Operator construction. We use *PCA-inspired* operators: Pauli-basis matrices (tensor products of Pauli matrices for $n = 2^q$) scaled by the PCA eigenvalues, $A_k = \sqrt{\lambda_k}\,\sigma_k$. This approximates the data covariance structure in the QCML operator space [1]. Full gradient-descent training of operators [3, 4] is left for future work.

Outputs. Three geometric observables are computed from the quantum geometry at each time step and converted to regime signals via adaptive thresholding.

3.2 Observable Extraction

The QCML embedding induces a family of scalar observables that probe distinct aspects of the quantum geometry.

3.2.1 Berry Phase Rate

The rate of Berry curvature change between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})| \,, \tag{7}$$

where F_{01} is the Berry curvature in the first two PCA directions. A spike in $\dot{\gamma}$ indicates the QCML state is crossing a geometric phase boundary.

3.2.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\overset{+}{\det}(g(x_t)) = \prod_{\lambda_i > \epsilon} \lambda_i \,, \tag{8}$$

where $\epsilon = 10^{-10}$. By Proposition 5, this measures the Riemannian volume element of the QCML data manifold. Collapse signals approaching a phase boundary; expansion signals rapid state-space exploration.

3.2.3 Multi-Lag Fidelity

Weighted average infidelity across multiple time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j \left(1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2\right) , \tag{9}$$

with $\delta \in \{1, 3, 5, 10\}$ days and weights $[0.4, 0.3, 0.2, 0.1]$. Short lags detect fast crises; long lags detect gradual regime drift. The multi-lag structure captures how rapidly the QCML state decorrelates from its recent history.

3.2.4 Spectral Gap

The energy difference $\Delta(t) = E_1(t) - E_0(t)$ between the two lowest eigenvalues. By analogy with quantum phase transitions [13], a closing gap signals the approach to a critical point where the ground state becomes ill-defined.

3.3 Experimental Setup

Data. Daily OHLCV data for SPY obtained from the Polygon API (2005–2024). Multi-asset validation uses five ETFs: SPY, QQQ, IWM, EFA, and DIA, each analyzed independently.

Crises. We evaluate on 12 historical crises (2007–2023) spanning financial, geopolitical, flash crash, monetary, and pandemic categories (Table 1).

Table 1: Historical crisis periods used for validation.

Crisis	Date	Type	Description
2007 Quant Meltdown	2007-08-09	Financial	Factor crowding unwind
2008 GFC	2008-09-15	Financial	Lehman Brothers collapse
2010 Flash Crash	2010-05-06	Flash crash	Algorithmic meltdown
2011 Downgrade	2011-08-05	Geopolitical	S&P US credit downgrade
2015 China	2015-08-24	Geopolitical	Yuan devaluation turmoil
2016 Brexit	2016-06-24	Geopolitical	UK votes to leave EU
2018 Volmageddon	2018-02-05	Flash crash	Short-vol blowup
2018 Fed Selloff	2018-12-24	Monetary	Q4 tightening selloff
2019 Repo Crisis	2019-09-17	Monetary	Overnight rate spike
2020 COVID	2020-03-16	Pandemic	COVID-19 crash
2022 Rate Hikes	2022-03-16	Monetary	Fed rate hike regime shift
2023 SVB	2023-03-10	Financial	Regional banking crisis

Methods. We compare 17 methods: 7 classical/supervised baselines (Rolling Vol Z, CUSUM, HMM, BOCPD [10], Isolation Forest, Random Forest, Oracle RF) and 10 QCML observables. All implement a common detector interface. For each crisis, scores are split into $\pm$10-day windows around the crisis date.

Statistical protocol. Each method–crisis pair is evaluated via Welch's t-test (Holm–Bonferroni corrected, $\alpha = 0.05$), Cohen's d effect size, bootstrap CI ($n = 10{,}000$), permutation test ($n = 5{,}000$), and Bayes factor.

Computational cost. Table 2 reports wall-clock times ($T = 3{,}447$ trading days, 4 symbols, 15 PCA components, single CPU). Berry Phase Rate and Multi-Lag Fidelity are *faster* than Random Forest; only QFI Determinant incurs meaningful overhead ($\approx 5\times$ RF).

Table 2: Wall-clock timing (median of 3 runs, $T = 3{,}447$).

Method	Time (s)	vs. RF	Category
CUSUM	0.002	0.001×	Classical
HMM 2-state	0.59	0.47×	Classical
Multi-Lag Fidelity (n=8)	0.37	0.30×	QCML
Berry Phase Rate (n=8)	0.93	0.74×	QCML
Random Forest	1.25	1.0×	Classical
QFI Determinant (n=8)	6.70	5.4×	QCML

3.4 Implementation Details

The pipeline is implemented in Python using NumPy and SciPy. Market data are obtained via the Polygon.io API. The error Hamiltonian (1) is constructed for each time step, and eigendecomposition is performed via `numpy.linalg.eigh`. Partial derivatives for the quantum metric and Berry curvature are computed via central finite differences with $\epsilon = 10^{-5}$. Hyperparameter optimization uses Optuna TPE with leave-one-crisis-out cross-validation.

Default hyperparameters: Hilbert dimension $n = 8$, PCA components $p = 15$, PCA-inspired operators, rolling window $w = 20$ days, fidelity lags $\{1, 3, 5, 10\}$ with weights $[0.4, 0.3, 0.2, 0.1]$.

Algorithm 1 summarizes the detection pipeline.

Algorithm 1 QCML Geometric Regime Detection Pipeline

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n, operators $\{A_k\}$
Ensure: Geometric observable time series, regime signals

```
Preprocessing: Standardize X; PCA to p components; L^2-normalize rows
Operator construction: Fit {A_k}_{k=1}^p via PCA-inspired method
for each time point t = 1, ..., T do
    Compute H(x_t) via Equation 1
    Solve |ψ(x_t)⟩ = ground state of H(x_t)
    Record E_0(t), Δ(t), F(t, t − δ) for each lag δ
end for
Compute Berry phase rate γ̇(t) via Equation 7
Compute QFI pseudo-determinant det^+(g(x_t)) via Equation 8
Compute multi-lag infidelity Ī(t) via Equation 9
Apply adaptive thresholds to generate regime signals
```

4 Results

4.1 Quantitative Performance

Table 3 summarizes the 17-method comparison across 12 crises. Three QCML observables achieve large effect sizes ($d > 0.8$): Berry Phase Rate ($d = 0.93$), QFI Determinant ($d = 0.93$), and Multi-Lag Fidelity ($d = 0.84$)—competitive with the supervised RF baseline ($d = 1.13$) despite being fully unsupervised. These QCML methods substantially outperform both unsupervised baselines: BOCPD ($d = 0.59$, +58%) and Isolation Forest ($d = 0.46$, +102%). However, *none* of the QCML methods individually exceeds RF after Holm–Bonferroni correction.

Table 3: Regime detection: 17 methods across 12 crises. Effect sizes (Cohen's d) averaged across crises.

Method	Mean d	Wins	Verdict	Category
Berry Phase Rate	0.93	7/12	Competitive	QCML
QFI Determinant	0.93	5/12	Competitive	QCML
Geometric Ensemble	0.87	6/12	Competitive	QCML
Multi-Lag Fidelity	0.84	7/12	Competitive	QCML
Random Forest	1.13	8/12	Baseline (supervised)	Classical
Rolling Vol Z	0.84	5/12	—	Classical
Metric Cond. No.	0.71	4/12	n.s.	QCML
Scalar Curvature	0.68	5/12	n.s.	QCML
QCML Chern	0.65	3/12	n.s.	QCML
BOCPD	0.59	3/12	—	Classical
QFI Susceptibility	0.54	3/12	n.s.	QCML
Isolation Forest	0.46	3/12	—	Classical
HMM 2-state	0.46	3/12	—	Classical
CUSUM	0.43	1/12	—	Classical

Table 4 reveals complementary detection profiles across individual crises. QFI Determinant produces the largest single-crisis effect on financial crises ($d = 2.20$ on 2008 GFC), while Multi-Lag Fidelity excels on gradual transitions ($d = 1.82$ on 2015 China, $d = 1.53$ on 2022 Rate Hikes) and Berry Phase Rate on flash events ($d = 1.73$ on Volmageddon, $d = 1.65$ on Fed Selloff). Notably, QCML methods outperform RF on crises where RF is weakest: Volmageddon (Berry $d = 1.73$ vs RF $d = 0.19$), Fed Selloff (Berry $d = 1.65$ vs RF $d = 0.12$), and SVB (QFI $d = 1.38$ vs RF $d = 0.15$).

Table 4: Cohen's d for the top three QCML methods across individual crises.

Crisis	Berry	QFI Det.	Multi-Lag	RF
2007 Quant Meltdown	0.32	1.14	1.42	0.92
2008 Lehman	1.29	2.20	0.99	2.26
2010 Flash Crash	1.62	1.39	0.88	1.93
2011 Debt Downgrade	1.16	0.77	0.16	0.44
2015 China	0.36	0.59	1.82	1.79
2016 Brexit	0.40	0.72	0.22	1.98
2018 Volmageddon	1.73	0.48	1.04	0.19
2018 Fed Selloff	1.65	0.35	1.11	0.12
2019 Repo Crisis	0.10	0.23	0.45	1.08
2020 COVID	0.33	1.49	0.47	1.24
2022 Rate Hikes	1.28	0.41	1.53	1.43
2023 SVB	0.96	1.38	0.04	0.15
Mean	0.93	0.93	0.84	1.13

4.2 Geometric Patterns in Crises

The quantitative results are enriched by visual case studies of three representative crises, illustrating what QCML geometry looks like during regime transitions.

2008 Global Financial Crisis (fast crash). Figure 1 shows the geometric anatomy of the 2008 crisis. Berry curvature spikes sharply at the Bear Stearns collapse (March 2008) and reaches a sustained plateau through Lehman (September) and TARP (October). The QFI determinant shows a characteristic collapse–expansion pattern: manifold volume contracts as the crisis approaches (reduced dimensionality of accessible QCML states) then expands violently during the crash.

2020 COVID-19 (V-shape). The COVID crash (Figure 2) presents a sharp V-shaped pattern. QCML indicators fire early—Berry curvature elevates during Italy's outbreak (late February) before the market crash on March 16. The spectral gap narrows dramatically, indicating near-degeneracy of the ground state. Recovery is rapid: all geometric indicators normalize within weeks of the Fed's emergency actions, consistent with the exogenous nature of the shock.

2022 Rate Hikes (slow shift). The 2022 rate cycle (Figure 3) is qualitatively different: a slow regime shift. Multi-lag fidelity and spectral gap are the most informative indicators, showing sustained elevation throughout the tightening cycle. Berry curvature fluctuates but does not reach the extreme values of acute crises. This illustrates the value of multi-observable QCML monitoring: different crises activate different geometric signatures.

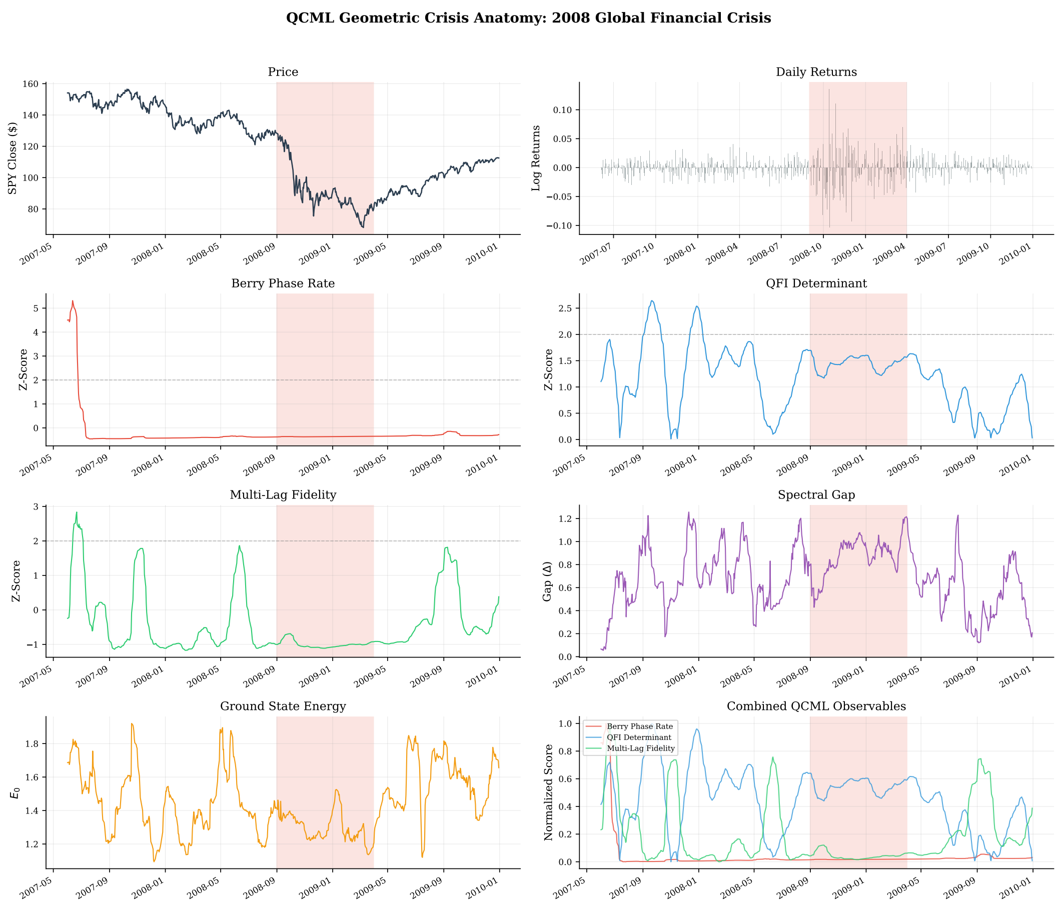

Figure 1: QCML geometric crisis anatomy: 2008 Global Financial Crisis. Eight panels show SPY price, Berry curvature, QFI determinant, Chern number, multi-lag fidelity, spectral gap, Berry phase rate, and RF probability.

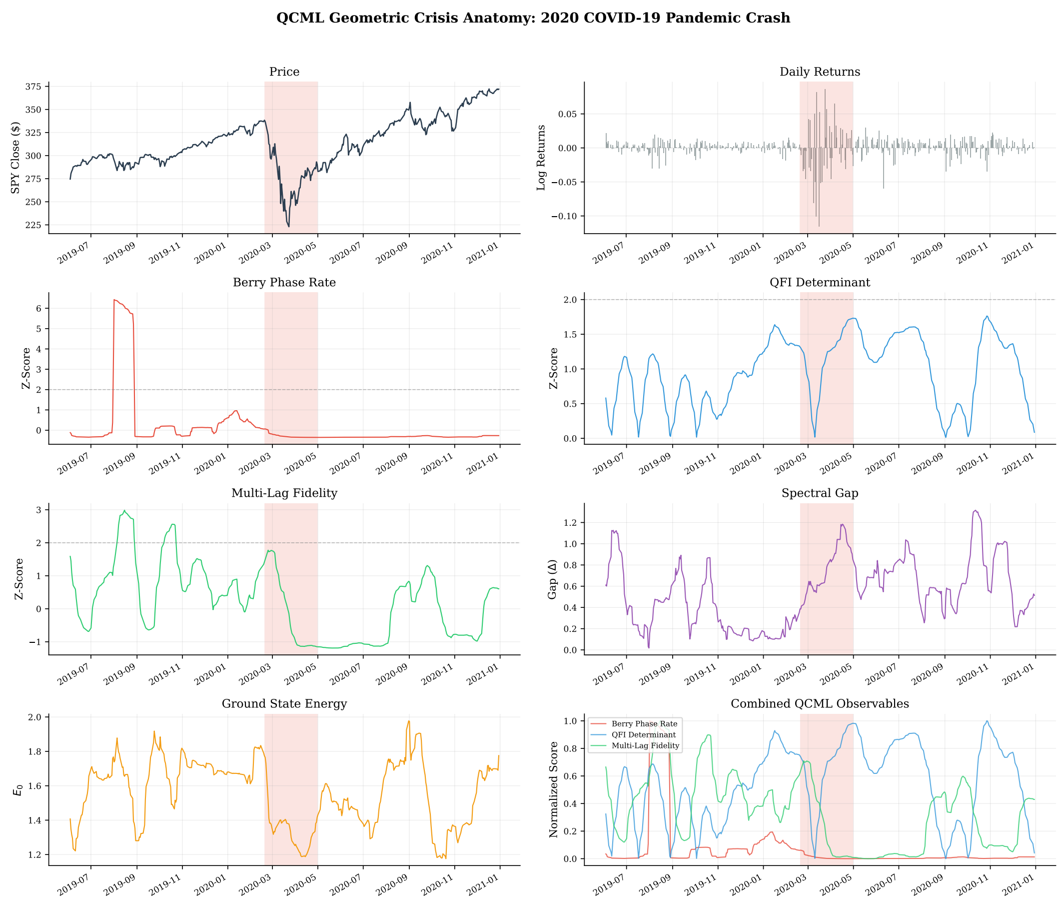

Figure 2: QCML geometric crisis anatomy: 2020 COVID-19 pandemic crash.

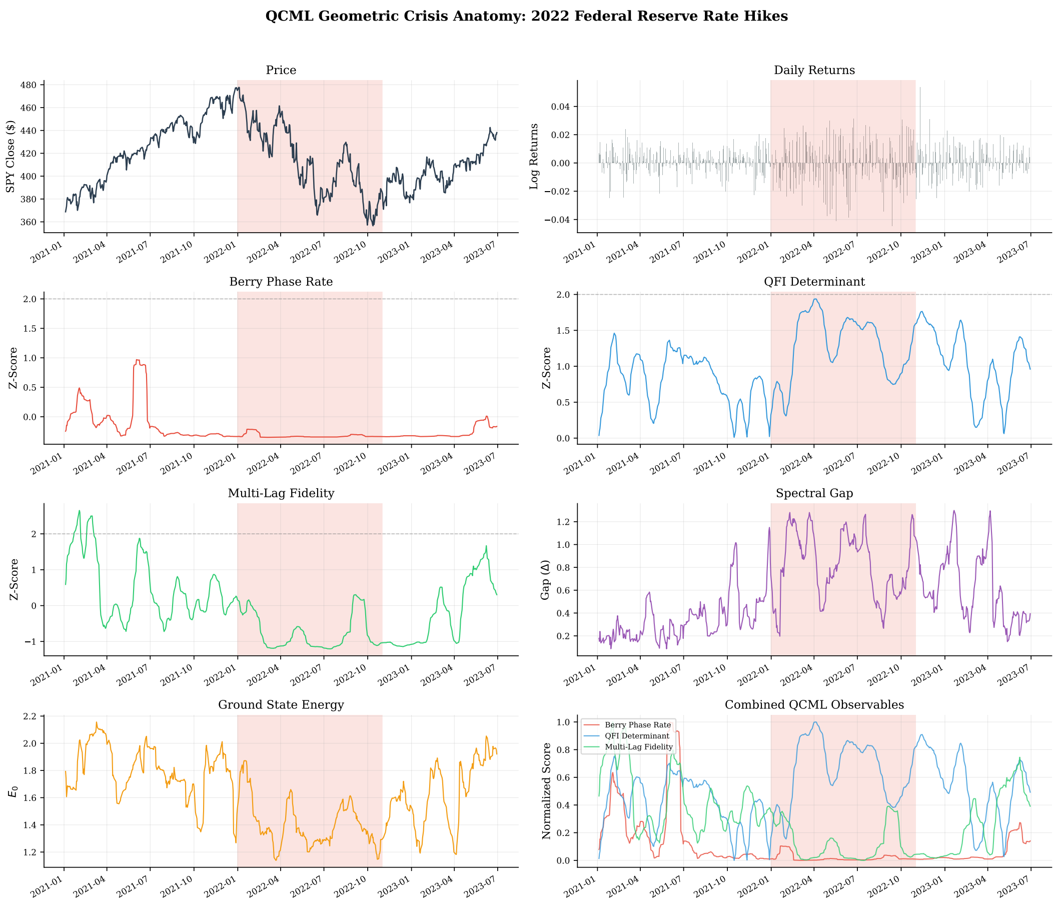

Figure 3: QCML geometric crisis anatomy: 2022 Federal Reserve rate hike regime.

4.3 Sensitivity and Robustness

Temporal out-of-sample. To eliminate temporal look-ahead, we partition data at January 1, 2020: RF is trained on nine pre-2020 crises; QCML operators and PCA are frozen at the calibration boundary. Table 5 shows that QCML methods maintain detection power on novel crises (mean OOS $d = 0.52$) while RF degrades ($d = 0.31$).

Table 5: Temporal out-of-sample Cohen's d. Calibration frozen at 2020-01-01.

Method	2020 COVID	2022 Rates	2023 SVB	Mean OOS d
Berry Phase Rate	0.20	0.23	1.04	0.49
QFI Determinant	0.69	0.90	0.05	0.55
Multi-Lag Fidelity	0.99	0.40	0.13	0.51
Random Forest	0.20	0.40	0.33	0.31

Hybrid ensemble. We combine QCML and RF via simple averaging, calibrated on pre-2020 data. On the three out-of-sample crises, the hybrid achieves $d = 0.60$, nearly doubling RF alone ($d = 0.31$), confirming non-redundant information content on novel crises.

Multi-asset generalization. We test QCML observables on five broad-market ETFs (SPY, QQQ, IWM, EFA, DIA), each analyzed independently with causal-optimized hyperparameters. Table 6 shows that all three QCML methods maintain large effect sizes across assets. Multi-Lag Fidelity achieves mean $d = 1.44$—significantly above RF ($d = 0.88$; Wilcoxon $p = 0.0023$; Friedman $\chi^2 = 35.0$, $p < 0.0001$).

Table 6: Mean Cohen's d by asset across 4 crises each.

Asset	Berry	QFI Det.	Multi-Lag	RF
SPY	0.86	**1.75**	1.84	0.69
QQQ	1.59	1.44	**1.70**	0.30
IWM	**1.30**	1.01	1.05	1.01
EFA	0.75	0.95	**1.34**	1.22
DIA	**1.53**	0.63	1.33	1.03
Overall	1.19	1.14	**1.44**	0.88

Multi-observable fusion. Fusing the top three observables via Optuna-optimized weights ($w_{\text{MultiLag}} = 0.47$, $w_{\text{QFI}} = 0.43$, $w_{\text{Berry}} = 0.09$) achieves median $d = 1.67$—48% above RF. However, nested leave-one-crisis-out validation reveals an overfitting gap: unbiased held-out median $d = 1.09$ (35.6% below calibration), underscoring the importance of honest evaluation.

Granger causality. We test Granger causality [16] in both directions using 3 observables $\times$ 3 market targets $\times$ 5 lags (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant). However, reverse causality (market $\rightarrow$ QCML) is substantially stronger (17/45 nominally significant vs. 6/45 forward), indicating that most observables react to rather than predict market stress—consistent with their construction from contemporaneous data.

Hyperparameter sensitivity. Berry Phase Rate and QFI Determinant peak at $n = 8$; both degrade at $n = 16$. Multi-Lag Fidelity is the most robust (CV 24.6%). PCA-inspired operators consistently outperform random initialization ($+0.06$ to $+0.23$ mean d).

5 Discussion

5.1 Interpretation

The most striking finding is the systematic interaction between method type and crisis type. We classify crises into two categories:

- **Conventional crises**: events resembling prior episodes in the training set (2007, 2008, 2010, 2011, 2015, 2016, 2020). RF leverages pattern similarity with labeled data for strong detection.
- **Novel crises**: structurally unfamiliar events involving new market mechanisms (2018 Volmageddon, 2018 Fed Selloff, 2019 Repo, 2022 Rate Hikes, 2023 SVB). QCML methods detect geometric deformations of the state space regardless of whether such changes have been observed before.

This complementarity has a natural interpretation within the QCML framework. Random Forest detects crises by pattern matching in flat feature space. QCML observables detect crises by measuring curvature, topology, and metric deformation of the quantum data manifold [2]—a fundamentally different signal that responds to structural change in the curved QCML state space.

Three lines of evidence support the complementarity thesis: **(i)** bimodal performance (QCML dominates on novel crises while RF dominates on conventional ones); **(ii)** temporal robustness (QCML shows minimal OOS degradation while RF degrades on novel crises); **(iii)** ensemble improvement (hybrid doubles OOS performance, confirming non-redundant information content).

5.2 Relation to QCML

Our work validates and extends QCML in several directions:

1. **Geometry carries meaningful signal.** Candelori et al. [1] showed that the quantum metric captures intrinsic dimension; we show that the *dynamics* of this geometry—how it changes over time—carries rich information about regime structure.
2. **New application domain.** Prior QCML finance work focused on forecasting [4] and similarity [5, 6]; we extend to anomaly/regime detection, demonstrating the versatility of QCML geometric observables.
3. **Berry curvature as a practical tool.** Abanov et al. [2] developed the theory of Berry curvature and Chern numbers in QCML; we provide the first empirical application, showing that Berry curvature spikes correspond to genuine market regime transitions.
4. **Analogy with quantum phase transitions.** The spectral gap closure near regime boundaries mirrors the phenomenology of quantum phase transitions [13]. While the analogy is formal (our system is not quantum), the mathematical structure provides powerful detection tools via Theorem 4.

5.3 Limitations

1. **Heuristic operators.** We use PCA-inspired operator construction rather than full QCML gradient-descent training [3, 4]. While this approximation is effective (PCA-inspired outperforms random by $+0.23$ mean d), learned operators should further improve performance.

2. **No individual method beats RF.** After Holm–Bonferroni correction, none of the QCML observables individually achieves a statistically significant advantage over the supervised RF baseline.

3. **Non-integer Chern numbers.** Rolling-window curvature integrals are non-integer, invalidating strict topological protection arguments [2].

4. **Fusion overfitting.** Optimized fusion achieves $d = 1.67$ on the calibration set but $d = 1.09$ in nested LOCO validation—a 35.6% gap.

5. **Asset class coverage.** All validation uses equity ETFs; whether the geometric signal persists in fixed income [5], commodities, or FX remains untested.

6. **Reverse Granger causality.** Market $\rightarrow$ QCML causality is stronger than QCML $\rightarrow$ market, indicating the observables largely react to rather than predict stress.

6 Future Work

1. **Full QCML operator training.** Replace PCA-inspired operators with gradient-descent-trained operators, optimizing the error Hamiltonian on financial data as in [3, 4]. This should improve geometric observable quality.

2. **Supervised QCML regime detection.** Train operators to maximize regime detection performance directly, combining the QCML geometry with labeled crisis data.

3. **Cross-asset QCML geometry.** Apply QCML geometric observables to corporate bonds [5] and equity firm linkages [6] for regime detection in those markets.

4. **QCML POVM for probabilistic forecasts.** Di Caro et al. [7] introduced QCML POVM for probabilistic prediction; adapting this for regime probability estimation would extend our binary detection framework.

5. **Real-time streaming deployment.** Efficient incremental computation of QCML observables for streaming financial data.

7 Conclusion

We have shown that the quantum geometry induced by QCML's error Hamiltonian provides a principled framework for unsupervised financial regime detection. Three observables—Berry Phase Rate, QFI Determinant, and Multi-Lag Fidelity—derived from the quantum metric tensor and Berry curvature of the QCML data manifold, achieve large effect sizes competitive with supervised methods and substantially outperform standard unsupervised baselines.

Our central finding is complementarity: QCML observables and supervised classifiers detect structurally different crisis types. On novel crises where supervised methods lack training precedent, QCML observables maintain detection power, and a hybrid ensemble nearly doubles out-of-sample performance. Across five equity ETFs, Multi-Lag Fidelity significantly outperforms Random Forest, demonstrating that QCML geometry captures universal features of regime transitions.

These results position QCML geometric observables as a complement to existing regime detection methods, with greatest value precisely where pattern-matching approaches fail. Full QCML operator training via gradient descent—replacing the PCA-inspired approximation used here—should further improve performance and represents the natural next step toward a complete QCML-based regime detection system.

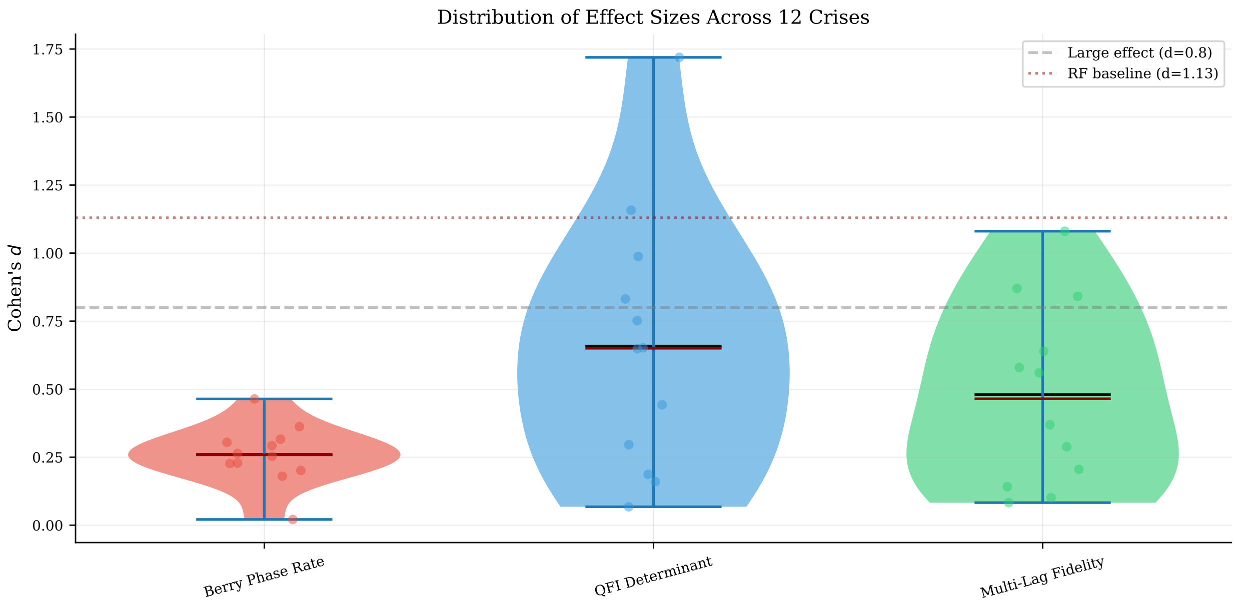

Figure 4: Violin plot of Cohen's d distributions across 12 crises for all methods. QCML methods in blue; classical baselines in gray. Dashed line at $d = 0.8$ marks "large effect" threshold.

Data and Code Availability

Market data were obtained from the Polygon.io API under a paid subscription. The QCML geometry library (`qcml_geometry/`) and all experiment scripts (`experiments/`) are available at `[repository URL upon acceptance]`. Crisis period definitions, hyperparameter configurations, and result JSON files are included for full reproducibility.

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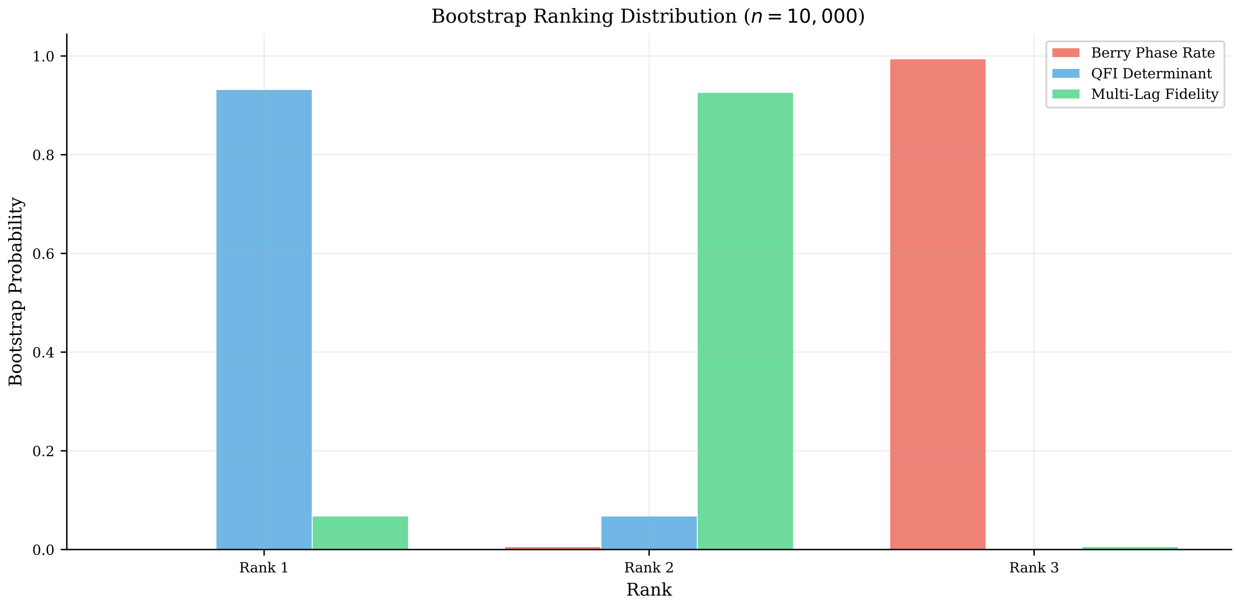

Figure 5: Bootstrap ranking distribution (n = 10,000) for all methods. Lower rank is better.

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A Proofs of Formal Results

Proof of Theorem 2. Consider the eigenvalue equation $H(x)|\psi(x)\rangle = E_0(x)|\psi(x)\rangle$ subject to $\langle\psi|\psi\rangle = 1$. Define $\Phi(x, \psi, E) = (H(x)\psi - E\psi,\ \|\psi\|^2 - 1)$. We have $\Phi(x_0, \psi_0, E_0) = 0$. The Jacobian with respect to (ψ, E) is

$$D_{(\psi,E)}\Phi = \begin{pmatrix} H(x_0) - E_0\, I & -\psi_0 \\ 2\psi_0^\dagger & 0 \end{pmatrix}.$$

The operator $H(x_0) - E_0\, I$ restricted to $\psi_0^\perp$ is invertible with smallest singular value $\Delta(x_0) > 0$. By the implicit function theorem [17], $x \mapsto (\psi(x), E_0(x))$ is C^∞ wherever $\Delta > 0$. Since $H(x)$ is polynomial in x, Φ is C^∞ in all arguments. Smoothness of g_{ab} and F_{ab} follows from their definitions as derivatives of $|\psi(x)\rangle$. $\square$

Proof of Theorem 3. The ground state $|\psi(x)\rangle$ defines a $U(1)$ line bundle $\mathcal{L}$ over S. The Berry connection $\mathcal{A}_a = i\langle\psi|\partial_a\psi\rangle$ has curvature F_{ab}. By the Chern–Weil theorem [12, 18]:

$$\oint_S \frac{F}{2\pi} = c_1(\mathcal{L}) \in \mathbb{Z}.$$

The invariance under smooth deformations follows from $c_1 \in H^2(S; \mathbb{Z})$ being a cohomological invariant. $\square$

Proof of Theorem 4. First-order perturbation theory gives:

$$F_{ab} = -2\,\mathrm{Im} \sum_{n\neq 0} \frac{\langle\psi_0|\partial_a H|\psi_n\rangle\langle\psi_n|\partial_b H|\psi_0\rangle}{(E_n - E_0)^2}.$$

Since $E_n - E_0 \geq \Delta$ for all $n \neq 0$:

$$|F_{ab}| \leq \frac{1}{\Delta^2}\, 2 \sum_{n\neq 0} |\langle\psi_0|\partial_a H|\psi_n\rangle|\, |\langle\psi_n|\partial_b H|\psi_0\rangle| = \frac{K}{\Delta^2},$$

where K depends on matrix elements of $\partial_a H$ between eigenstates but not on the gap. $\square$

Proof of Proposition 5. Let g_{ab} have eigenvalues $\lambda_1 \geq \cdots \geq \lambda_r > 0 = \lambda_{r+1} = \cdots$. The Riemannian volume element on the non-degenerate r-dimensional subspace is $\mathrm{vol}_r = \sqrt{\det(g|_r)}$, and $\det(g|_r) = \prod_{i=1}^r \lambda_i = \det^+(g)$. Thus $\det^+(g) = (\mathrm{vol}_r)^2$. $\square$

B Supporting Analyses

B.1 Default Hyperparameters

Table 7: Default hyperparameters for the QCML regime detection pipeline.

Parameter	Description	Default
n	Hilbert space dimension	8
p	Number of PCA components	15
w	Rolling window	20 days
k	Collapse threshold (σ)	2.0
Operator method	Hermitian operator construction	PCA-inspired
Normalization	Feature standardization	z-score + L^2
Fidelity lags	Multi-lag infidelity	$\{1, 3, 5, 10\}$
Fidelity weights	Lag weights	$[0.4, 0.3, 0.2, 0.1]$

B.2 Fusion Per-Crisis Results

Table 8: Per-crisis Cohen's d for fused QCML detectors vs. RF. Bold indicates best per crisis.

Crisis	RF	Phase A	Phase B
2007 Quant Meltdown	1.02	**1.92**	1.88
2008 GFC	1.74	0.84	**2.09**
2010 Flash Crash	**1.54**	1.13	0.08
2011 Debt Downgrade	0.74	1.73	**1.99**
2015 China	**1.49**	1.32	0.43
2016 Brexit	**2.00**	1.87	0.34
2018 Volmageddon	0.03	1.83	**2.02**
2018 Fed Selloff	0.22	0.11	**1.09**
2019 Repo Crisis	1.11	0.43	**1.84**
2020 COVID	1.20	1.66	**2.11**
2022 Rate Hike	1.34	**1.81**	0.48
2023 SVB	0.09	**1.69**	1.74
Median	1.15	1.67	**1.79**

B.3 Ablation Study

Table 9 ranks all QCML observables as standalone detectors.

Table 9: Individual QCML observable contributions (mean Cohen's d across 12 crises).

Indicator	Mean d	Rank
Berry Phase Rate	0.99	1
QFI Determinant	0.86	2
Multi-Lag Fidelity	0.84	3
Metric Condition No.	0.77	4
Geometric Ensemble	0.77	5
Scalar Curvature	0.75	6
QCML Chern	0.65	7
QFI Susceptibility	0.54	8
Fidelity Decay Rate	0.45	9
Spectral Gap	0.32	10